Rebuilding America's Industrial Middle

Initial takeaways on AI-native, vertically integrated industries

1. Key takeaways

1. **Some of the forces pulling production back to the US are temporary and others are lasting. The most durable opportunities for us are companies that own production in the parts of the chain that will stay short: processed materials, skilled labor and qualified capacity.**
   - **Temporary forces:** the reshoring push rides on one policy lever after another: Section 232 metals tariffs, pauses in China's export controls, CHIPS grants (FY2026 was the last funded year) and a munitions budget surge whose funding is still pending. Each can be reversed by a proclamation, a trade deal or a budget vote. Companies whose economics only work behind one of these are priced at a premium while it lasts and may lose their edge when it ends.
   - **Lasting limits:** three things are different. China controls the middle of the supply chain (refining, magnets, chemical building blocks), and building alternative plants takes years. The workforce that makes things is aging out: the US needs 3.8M more manufacturing workers by 2033 and about 1.9M of those jobs may go unfilled (Deloitte / Manufacturing Institute). And buyers take years to qualify a new supplier, so qualified capacity cannot be added quickly.
   - **Implications:** we should favor companies that own production where these limits bite and would still win if tariffs came off: more output per skilled worker, faster qualification and plants outside China. These companies benefit as the limits tighten, without needing policy to stay where it is.
2. **China's hold on critical materials is in processing, not in the ground. The value sits with whoever owns a working plant outside China and can sign long-term supply contracts, not with whoever finds the deposit.**
   - China is the leading refiner of 19 of 20 strategic energy minerals, at about 70% average share (IEA), and makes over 90% of rare-earth magnets while the US makes under 1%. The US imports all of its supply of 16 mineral commodities (USGS). Ore exists in many countries; what is missing are the plants that separate and refine it and turn it into metal and magnets.
   - China uses that position. It licenses exports of heavy rare earths and magnets, has paused broader controls only until November 2026, and can cut prices to make Western plants uneconomic.
   - In response the government now lends, invests and buys: Pentagon loans through the Office of Strategic Capital ($620M to Vulcan Elements), about $27B of equity stakes across some 30 companies, and a 10-year price floor for MP Materials' rare-earth oxide. Government lenders want a long-term supply contract, which only the owner of a plant can sign.
   - Explorers and miners do not own these economics. They sell a deposit or a commodity at a price set globally or in China, and a deposit earns nothing until a mine is built, on average 15.7 years after discovery. We would back processing that already has a government or defense buyer, not exploration.
3. **The bottleneck in US manufacturing is a small, aging and under-capitalized supplier base, not a lack of machines. Companies that get more output from each skilled worker fill the gap that defense and reshoring demand is exposing.**
   - Manufacturing is a pyramid. A handful of prime contractors sit at the top (51 shrank to 5 after the Cold War), with thousands of small shops below them making parts, castings and forgings. 83% of the 17,156 US machine shops have fewer than 20 employees, and many owners are near retirement with no successor.
   - Existing machines are underused: the average CNC spindle cuts metal 34% of the time against a 70% target. The shortage is people, programming and process, not equipment.
   - Labor cost is not the issue. In a new automated plant labor is under 10% of the cost of goods (Drew Baglino); China wins on a supply chain within a three-hour drive, not on wages. Software that automates

quoting, programming, scheduling and inspection lets each worker run more machines: Hadrian trains new operators in 30–40 days instead of years.

- So the pitch that works is “we deliver when your supplier cannot hire”, not “we are cheaper”.

4. **Defense is the largest source of demand, but it is lumpy and gated by qualification. Buyers pay new suppliers for speed and capacity rather than lower prices, and the prize is turning prototype work into multi-year orders.**
   - The FY27 munitions request is $49.5B, 8.5x the prior decade’s average, but 82% of it depends on a reconciliation bill, and orders lag budgets: a specialty-alloy maker’s defense bookings fell because its customers had not received funds (CNAS; Carpenter). Primes are already signing capacity deals with new producers, such as Lockheed’s seven-year agreement to put Hadrian cells inside its missile plant.
   - Most buyers do not value speed alone; they value scale, reliability and total program cost (Hadrian’s CEO). Where a delay is costly they pay a premium: a Northrop engineer would pay 20–40% more to avoid one.
   - Qualifying a new supplier takes one to several years, and buyers cap any one supplier at about a third of a part family. Once paid, qualification locks a supplier into programs that last decades. The FY26 defense bill adds multiyear munitions buys and purchase commitments with advance payments, the first channels that can turn prototypes into repeat orders.
5. **Plants have to be built before contracts arrive, so the capital structure decides who survives. Equity should pay for software and the first plant, and debt and government money for the rest, which favors capital-light models at our stage.**
   - Defense buyers will not sign a production contract until the facility exists, so spending on plants runs about two years ahead of revenue (Chris Power, Hadrian). Building has also become harder: the Fed raised rates to 3.75–4.00%, tariffs raise input costs, and factory construction spending is down about 20% from a year ago.
   - The founders who have scaled say equity should never fund factory construction: Hadrian builds plants with a bank credit line and Navy money, and Vulcan with a Pentagon loan. Lenders will fund machines, but not software teams.
   - Venture money is concentrating: later-stage industrial funding grew about 10x in four years, and the leaders are priced like AI companies (Hadrian at $7.87B). The models that fit our check grow without a megafactory: automating existing plants, franchising, buying certified shops, or small plants funded with equipment debt.
6. **Data that only exists by making things is the lasting advantage, because AI models themselves are becoming free. The claim that owning production makes each unit cheaper or better over time is still mostly unproven.**
   - Manufacturing data is “never on the internet” (Chris Power), and published materials data is noisy and leaves out failed experiments (Periodic Labs). A company that runs its own plant or lab produces data no software vendor sees.
   - The models are commoditizing: Google DeepMind’s GNoME, Microsoft’s MatterGen and the Department of Energy’s Genesis Mission give everyone free models, data and compute. The scarce assets are physical throughput, proprietary experimental data, and the ability to scale up a product and get it qualified.
   - Software sold into other companies’ plants rarely sticks: Earth AI’s 80 pilots with miners did not recur, and GrayMatter switched to renting robot cells after software-only sales did not work. The owner of the operation keeps both the data and the margin.
   - The only learning-over-time number in our research is Hadrian’s 50% cut in time to first part. In chemicals and materials, discovery is fast but scale-up and customer qualification take one to five years each.
7. **The key risk for any new producer is that incumbents automate with the same tools while keeping their approvals. The most defensible companies own the plant, the software and the qualification together.**

- Incumbents use AI too: Foxconn reports 15% higher labor productivity from its NVIDIA-based factory agent, Northrop claims about 20% lower recurring cost at a new plant, and Carpenter says AI productivity gains are ahead of plan. Permanent 100% bonus depreciation makes new robots cheaper for profitable incumbents.
- Vendors sell the same stack to everyone: NVIDIA's factory software, CloudNC's AI machining software in 1,000+ shops including Lockheed's, and robot cells rented to legacy plants. a16z notes incumbents are building their own "robotic arm farms" for data.
- A company that automates one step can be copied. One that holds the qualification, the software and the plant together is much harder to replace.

2. Where we would invest

The areas below follow the five industries of the VII primer, in its order, which is also the order of the takeaways in Section 1: Raw Materials, Manufacturing, Specialty Chemicals, Advanced Materials, and Enabling Software & Robotics. Each opens with the takeaways it acts on and our view, then goes sub-group by sub-group. Every company carries one of three labels: fully integrated (owns production and sells the product; core fit), hybrid (owns some production but also sells services, licenses or relies on partners; partial fit) or enabler (sells tools to other companies' plants; outside the core).

A. Raw Materials (Minerals & Metals)

Our view: Selective. Back refining and magnets (Stage B) where a government or defense buyer is already paying. Conviction: medium.

How it ties to takeaways 2 and 5: takeaway 2 says the value in critical materials sits with whoever owns a working plant outside China, and takeaway 5 says government debt should fund the plant. We would invest only in Stage B, refining minerals and metals. Stage A, locating and mining, is excluded: explorers do not own the economics and are price-takers (Section 3).

Policy money lands here first: $65B of public money was committed to minerals in 2025, and DoD set a 10-year price floor with MP Materials. MP is still loss-making, so profits depend on prices a startup cannot control.

Stage B: Refining minerals/metals

Attractive now, selectively. The scarce asset is a permitted, working plant outside China. Mariana Minerals is the benchmark (one of two credible data loops), but its check size is above ours.

Vulcan Elements *(Fully integrated; $65M Series A, Altimeter; $620M OSC loan and $50M Commerce equity in a $1.4B package)*

- **Overview and use cases:** Vulcan makes sintered rare-earth magnets for drone and robot motors, chip-fab tools and cars from non-Chinese material in North Carolina. A 10-ton pilot runs in Durham, and a 10,000-ton plant is being built in Benson.
- **Thesis:** China makes over 90% of rare-earth magnets and the US under 1%, so a domestic producer has the government as buyer of last resort (takeaway 2). The Army chose Vulcan for its million-drones-a-year program, and the company says pilot output is sold out 12 months ahead. Government debt funds the plant (takeaway 5).
- **Areas to diligence:** The key question is whether unit cost gets close enough to China that commercial buyers sign without subsidy; no commercial buyer is named. Oxide supply from ReElement and Ucore arrives in 2027, and the CEO says the equipment is Chinese too. Its Form D shows about $114M of a $545M offering unsold, so we would check round timing and allocation.

Phoenix Tailings *(Fully integrated; $116.6M Series B; $500M conditional OSC loan; selected for a $66M DOE grant)*

- **Overview and use cases:** Phoenix turns rare-earth concentrate and recycled magnet scrap into the metals and alloys that magnet makers melt (NdPr, dysprosium, terbium), using a lower-energy route first funded by ARPA-E. It runs a 200 t/yr metal plant in New Hampshire, is building a $375M separation plant in Rochester, NY to ramp from 2027, and bought an AI and automation team in April 2026.
- **Thesis:** Phoenix is the most advanced US rare-earth metal maker, with a $100M contract from an undisclosed automaker and backing from BMW i Ventures, Yamaha and Sumitomo. It sits between oxide and magnet, the step where China's lead is greatest (takeaway 2).
- **Areas to diligence:** We would test a roughly 15x ramp, from 200 t to about 3,000 t of metal by 2028; the OSC loan and the DOE grant are both still conditional. Long-term feedstock is not locked in, customers are unnamed, and China sets NdPr prices with no public price floor like MP's. The check size is large.

B. Manufacturing

Our view: Selective. Back capital-light owner-operators where speed and qualification are already paid for; use the large names as benchmarks. Conviction: medium.

How it ties to takeaways 3, 4 and 5: takeaway 3 says the gap is skilled people, not machines, takeaway 4 says buyers pay for speed and qualified capacity, and takeaway 5 says capital-light producers fit our check. This is the most commercial industry in the primer: metal supply and fabrication already sell, and precision has revenue but mostly prototype and second-source defense volume.

Job shops lack capital, software and successors, and online marketplaces never own the margin. SendCutSend (Series B at $2.1B, profitable for eight years) shows the model works, and the top of precision is priced out (Hadrian $7.87B, Divergent $2.3B, Amca over $1B). We use those as benchmarks; our entry is Series B or a capital-light structure.

Metal Supply

Watch, not invest yet. Tariffs help US metal, but the margin sits with incumbents: Reliance is taking share without AI.

- **Nox Metals (fully integrated; $11.5M seed, Hyperion; we passed):** an automated aluminum and steel distributor outside Detroit with instant online quotes and nesting software, cutting quote time from 3–5 days to minutes (company claim). We would revisit if an AI-native distributor shows better margins or turns than Reliance at scale.

Fabrication (sheet/structures)

Attractive now. The cheapest lane to enter, and high machine use comes from scheduling software rather than robots.

Senra Systems *(Fully integrated; $65M Series B, July 2026, co-led by Lowercarbon and Interlagos; ~$493M post per Harmonic; ~190 employees)*

- **Overview and use cases:** Senra builds wire harnesses and cable assemblies for aerospace, defense and space in two of its own California plants (about 95,000 sq ft). Harnesses can be about 25% of a vehicle's cost and are still built almost entirely by hand. Its in-house platform, Amp, runs quoting, engineering, work instructions, supply chain, machine programming and quality in one system. The CEO ran SpaceX's harness organization.
- **Thesis:** Senra acts on takeaway 4: it sells speed (lead times from six months to two weeks, a company claim) on a part that gates vehicle production. Growth comes from existing customers, and every part it quotes or builds feeds Amp's library, the start of a data loop (takeaway 6). The Series B funds a third factory, and the stage fits us.

- **Areas to diligence:** The physical build is still manual on standard equipment, so today's edge is software plus trained labor, not robots; we would test whether the new AI and robotics push lowers cost per harness. 2026 sources do not confirm AS9100 or ITAR status. We would ask for first-pass yield (99% claimed), repeat-order data, gross margin and customer concentration.

1872 *(Fully integrated; $15M seed via Ohio's O.H.I.O. Fund; ~18 employees)*

- **Overview and use cases:** 1872 makes welded steel skids, frames and enclosures for energy and utility customers in one 30,000 sq ft plant in Cincinnati, using robots and its own software.
- **Thesis:** Data-center and grid buildout is pulling on fabricated enclosures (one legacy fabricator has backlogs into 2027), and 1872 is a "factory as the product" bet at a seed price. At this stage it is a relationship to build ahead of a second plant.
- **Areas to diligence:** Its welding AI belongs to Path Robotics, not 1872, so the software edge is unproven, and we found no customer evidence yet. We would wait for plant-one utilization, gross margin and a first repeat order.

Precision & Defense Hardware (machining, 3D printing, castings, composites)

Selective; mostly priced out at the top. Defense programs lock in suppliers for decades, but buyers cap any one supplier at about a third.

Freeform *(Fully integrated; $67M Series B, February 2026; ~$179M post per PitchBook; Founders Fund, NVIDIA, Boeing)*

- **Overview and use cases:** Freeform 3D-prints metal parts (heat exchangers, manifolds, rocket-engine injectors and nozzles) on its own multi-laser printers and sells parts, not printers. Cameras watch the molten metal at microsecond scale and machine-learning models adjust each layer live on an on-site GPU cluster. The founder ran SpaceX's Merlin and Raptor engine programs, and it won a DIU contract in July 2026.
- **Thesis:** Freeform has the clearest data loop in the category (takeaway 6): its process runs continuously and produces print data no vendor sees, which is why NVIDIA and Boeing invested. Selling parts rather than machines keeps the margin, and a valuation under $200M leaves an entry point.
- **Areas to diligence:** Customers are undisclosed, so we would ask for named references and whether any parts fly on a production program. We would ask for yield and cost-per-part trends over time, the evidence no one in the category has shown. We would also check capital needs for its larger printer and the co-founder's departure; its Form D shows about $8M of the round unsold.

Isembard *(Hybrid; $50M Series A, March 2026, led by USV; ~$250M per Harmonic estimate; 14 factories)*

- **Overview and use cases:** Isembard runs a network of precision machining factories (CNC milling and turning in aluminium, steel, titanium and Inconel) for customers such as Anduril, Babcock and Tekever. Every site runs its MasonOS software; franchisees bring the machines and building, and Isembard brings the software and the demand.
- **Thesis:** Isembard is a capital-light owner-operator in a qualification-gated market (takeaways 4 and 5). It reached 14 factories (4 owned, 10 franchised) in about 18 months, says revenue grew 10x year on year, and cut one customer's lead time from about 10 weeks to under 14 days at a matched price.
- **Areas to diligence:** We would test economics per franchise and how primes audit franchised sites, since a Thales buyer is skeptical. We would confirm AS9100 and ITAR coverage across the network and the plan and structure for US expansion from a UK base.

C. Specialty Chemicals

Our view: Selective. Of the names in the primer, the two US- and UK-based automated chemistry labs are the most interesting to us. Conviction: medium.

How it ties to takeaways 2 and 6: takeaway 2 says China controls the middle of the supply chain, which for pharma means the chemical building blocks, and takeaway 6 says data from running your own operation is the lasting advantage. The companies here run their own automated labs, make molecules for pharma and industry, and keep every experiment, including failures, as training data.

About 46% of US generic doses depend on Chinese source materials, and biotechs have long outsourced molecule-making to Chinese contractors such as WuXi AppTec. The BIOSECURE Act (enacted December 2025; first list of barred companies due December 2026) gives buyers a reason to move that work. The primer's other names are context rather than targets: Solugen is late-stage, Phlow waits on FDA approval in 2027, Osmo sells mainly to small fragrance brands, Scimplify is an Indian asset-light competitor to reshoring, and Enko faces 10–15-year pesticide registration.

Warp Sciences *(Hybrid; $215M, October 2026, led by Lightspeed at an $865M valuation; IVP, Premji Invest, Lux, Lowercarbon)*

- **Overview and use cases:** Warp is a US chemistry contract research organization (CRO) that combines expert chemists, AI and robotics to make the molecules biotech and pharma teams test. Its first service is lead-optimization analog synthesis: making variations of a promising molecule so customers can compare them. Its labs are designed for automation from day one; robots learn tasks from chemists wearing cameras and sensor gloves, and every step is logged as training data. CEO Kevin Weil ran product and science at OpenAI; COO DJ Patil was the first US Chief Data Scientist; CSO Sol Reisberg spent about 20 years in medicinal chemistry at Gilead and Schrödinger.
- **Thesis:** Warp is a direct bet on bringing offshored chemistry back to the US (takeaway 2): it aims to compete on speed, cost and IP protection as US–China tensions and pharma's IP concerns push buyers away from Chinese contractors. Because it owns the lab, it keeps experimental data no software vendor sees, the chemistry version of the plant-owner data loop (takeaway 6). Lowercarbon, which co-led Senra's Series B, is also an investor.
- **Areas to diligence:** Warp launched on October 22, 2026, and no customers, prices or throughput are public. US chemists cost far more than offshore CRO labor, so we would test how fast robots take over reaction set-up and purification and whether cost per compound actually falls. It sells research-scale services, not product, and at $865M on launch there may be no entry point now; we would track it toward the next round.

Chemify *(Hybrid; >$50M Series B, October 2025, Wing and Insight, with 8VC; £22M in UK and Scottish grants)*

- **Overview and use cases:** Chemify grew out of Lee Cronin's University of Glasgow lab in 2022. Its "Chemifarm" turns code written in a chemical programming language into molecules made by robots; Cronin calls it "AWS for chemistry". The first Chemifarm opened in Glasgow in June 2025, and an £89.9M expansion adds a second-generation Chemifarm in Govan and takes staff from 152 to about 450 in three years.
- **Thesis:** Chemify is the most ambitious "code to molecule" platform, and it earns in three ways: customers pay for molecules, for software access, and through co-owned IP. Cronin pitches distributed Chemifarms for supply-chain security (takeaway 2), and each run feeds its own data (takeaway 6). It says it has 12 customers and aims for 20 large ones (company claim, on a friendly podcast).
- **Areas to diligence:** Today it is a research-scale CRO and co-discovery shop, not a bulk producer, and its customers are unnamed. Co-owned IP can make revenue lumpy, and the expansion leans on government grants. We would test cost per molecule against offshore CROs, how much revenue comes from each of the three lines, and Cronin's stated goal of reaching positive cash flow within one venture cycle.

D. Advanced Materials

Our view: Selective. Back producers with an urgent buyer; watch the discovery labs. Conviction: low to medium.

How it ties to takeaways 4 and 6: takeaway 4 says qualification is the gate, and takeaway 6 says discovery is getting cheap while scale-up is not. The producers here make engineered materials (cooling fluids, battery cathodes) for buyers with a new need that no incumbent is qualified to meet.

China makes about 85% of cathode material and over 98% of the cheaper LFP cathode, and data-center power and EU PFAS rules create demand for new cooling materials. No AI-designed material sells at volume yet.

Producers

Selective; Orbital first.

Orbital Industries *(Fully integrated; $50M Series B, May 2026, led by Plural; ~50 employees)*

- **Overview and use cases:** Orbital sells an AI-designed, PFAS-free fluid for cooling data-center servers, made by a contract manufacturer, plus modular data centers. AWS is a strategic partner, and BASF licenses its design software.
- **Thesis:** Orbital owns the product margin and has an urgent buyer. Liquid cooling is expected to reach 53% of AI chips in 2026, and the EU's PFAS restriction pushes demand for replacements. If the fluid ships in 2027, it could be among the first AI-designed materials sold at scale.
- **Areas to diligence:** The fluid is in qualification for 2027 GPUs and none has been sold. We would ask who the contract maker is, gross margin per liter, and how much revenue today is software licensing rather than fluid.

Mitra Chem *(Fully integrated; Series B led by GM; $100M DOE grant and $50M from Michigan)*

- **Overview and use cases:** Mitra makes iron-phosphate cathode powders (LMFP and LFP), the active material in EV and grid batteries, and has shipped samples since 2023. It uses AI to take a chemistry from lab scale (1 kg a day) to production (1,000 kg a day) and plans a 30,000-ton, $370M LFP plant in Muskegon, Michigan.
- **Thesis:** Mitra sells security of supply into the chemistry the US makes least: more than 50 GWh of US cell capacity has moved to LFP, largely for grid storage. It is an AI-designed chemistry turning into a real plant, which passes our producer test.
- **Areas to diligence:** It competes on cost per kWh with Chinese LFP, the hardest market in industry, and its value depends on the 45X credit and foreign-entity rules. Customers in qualification are unnamed, and plant commissioning has slipped to 2028.

Materials Discovery Labs

Watch, not invest yet. Labs invent materials with AI and robots but none sells material at scale; they are priced on talent.

- **Radical AI (hybrid; $55M led by RTX Ventures):** invents hypersonic and propulsion alloys in a robot-run lab, paid for R&D by RTX and the Air Force; its RAI-939 alloy beat C103 in an independent torch test. It is the clearest defense-led path, but its contracts total about $1.2M and the jump from a 500 g sample to qualified mill product needs a melting and forging partner. We would revisit when a lab sells material or signs a supply contract.

E. Enabling Software & Robotics (tracked, outside the core)

Our view: Watch, not invest yet; track as benchmarks and as tools inside producers. Conviction: medium.

How it ties to takeaways 6 and 7: takeaway 6 says software sold into other companies' plants rarely sticks, and takeaway 7 says these tools arm incumbents as much as startups. We would consider at most one exception: software that is paid like a producer and runs the plant itself rather than advising an operator.

Traction is first-customer stage (Arda: Anduril only; Mind Robotics: Rivian), and GrayMatter moved to renting robot cells after software-only sales did not work. Siemens and Rockwell own the installed base.

Lumaril *(Enabler; $12M seed led by Lux; WSJ Pro reports $66M at a $400M valuation)*

- **Overview and use cases:** Lumaril connects to a plant's existing controls, adds cheap audio and video sensors, builds a real-time model and changes setpoints itself through a safety layer to raise throughput and yield in chemicals, biologics and polymers. Its founders created Palantir Foundry, and it says "we're paid on the production we add."
- **Thesis:** Lumaril earns a producer's margin and data loop without owning the plant, which lets us back the thesis in process industries where a new plant costs hundreds of millions. Chemical plants need 80–85% utilization to earn their cost of capital and Europe averages about 74%, so each extra point of yield in a sold-out plant is close to pure margin.
- **Areas to diligence:** No customers or results are public, so we would ask for one plant's before-and-after yield, how the pricing baseline is set, and how the plant's safety team approved closed-loop control. We would also assess competition from Honeywell, AspenTech and Emerson, and the owner's ability to end the contract. This is our lowest-priority area and would be at most one investment.

3. Where we would not invest

We would avoid layers where returns depend on commodity prices, government programs or being priced as an AI company, rather than on the go-to-market and customer support we provide.

- **Raw Materials, Stage A: locating and mining critical minerals (Earth AI, KoBold):**explorers do not own the economics. They sell deposits to miners or commodities at prices set globally or in China, and a deposit earns nothing until a mine is built, on average 15.7 years after discovery (19.1 in the US).
- **Raw Materials: equity-funded mines and greenfield refineries (Mariana-scale):**one large bet better suited to debt, with commodity prices that swing (lithium fell 24% in 2025; Ganfeng lost over 75% of its value).
- **Manufacturing: precision parts at the top (Hadrian, Divergent, Amca):**$1B–$7.9B valuations ahead of audited revenue; useful as benchmarks.
- **Specialty Chemicals: bio-based commodity chemicals and agrochemical actives:**Ginkgo, Zymergen and Amyris collapsed in 2021–23, and new crop actives take 10–15 years and about $300M to register.
- **Advanced Materials: discovery labs (Periodic, Lila, CuspAI):**priced on talent and none sells material yet; faster discovery does not shorten customer qualification.
- **Enabling Software & Robotics: factory software and robots sold to others (Arda, Mind Robotics):**first-customer stage, Siemens and Rockwell own the installed base, and Mind has raised over $1B with no public product.

4. What we would underwrite

We would look for companies that own the plant and its software, are already paid for speed or qualification, finance machines with debt, and can show each unit getting better or cheaper over time.

- **Owns production and the software that runs it,** so the margin and the data stay in-house.
- **Already paid for speed or qualification:** a named buyer, a repeat purchase order or a qualified part.
- **Debt or government money funds the machines;** equity pays for software and the first plant.
- **Improves with use:** scrap, cycle time, yield or quote accuracy improving with volume, or a plan to measure it.
- **Wins without policy:** would still sell if tariffs came off or a US–China deal lowered prices.
- **Better than job-shop margins,**above the roughly 30% gross margin of incumbent roll-ups and toward Proto Labs' 44.5%.